# Group Coursework Submission Form

## Specialist Masters Programme

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|---|---|
| **Please list all names of group members:**<br>1. Coulter, Matthew<br>2. Jain, Seher<br>3. Jaiswal, Sidhant | 4. Pinto, Aaron Neil<br><br>**GROUP NUMBER:** **10** |
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| **Lecturer: Dr** Vali Asimit | **Submission Date:** 16th July 2025 |

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# Advanced Predictive Analytics

## Submitted by

Matthew Coulter
Seher Jain
Sidhant Jaiswal
Aaron Neil Pinto

## Submission Date

July 16, 2025

**MSc Actuarial Science**

# Table of Contents

**Coursework 1: Shrinkage Estimation for Portfolio Construction** **4**
A1: Data Processing . . . 4
A2: Selection of Estimation Models . . . 6
Classical Approach: Sample Mean and Sample Covariance (Sample-Sample) . 6
D-Multi-Shrinkage Mean with Ledoit-Wolf Covariance Estimator (DM–cov1Para) 6
Oracle Linear Shrinkage Mean with Quasi-Inverse Shrinkage Covariance Estimator (OL–QIS) . . . 6
A3: Portfolio Construction and Rolling Window Setup . . . 7
A4: Portfolio Performance Evaluation . . . 9
Annualised Sharpe Ratios by Rolling Window . . . 9
Nominal Gains Ratios by Rolling Window . . . 10
Summary Statistics of Annualised Sharpe Ratios . . . 10
Summary Statistics of Nominal Gains (£) . . . 11
A5: Interpretation of Results and Executive Summary . . . 13
Risk-Adjusted Performance (Sharpe Ratios) . . . 13
Absolute Performance (Nominal Returns) . . . 13
Contextual Interpretation . . . 13
Final Takeaway . . . 14

**Coursework 2: Shrinkage Generalised Linear Models for Insurance Claims Estimation** **15**
B1: In-Sample Performance Evaluation . . . 15
B2: Out-of-Sample Performance Evaluation . . . 18

# Shrinkage Estimation for Portfolio Construction

## A1: Data Processing

In this initial step of the portfolio construction, we preprocess the provided financial data to compute excess returns for 100 selected U.S. companies. The analysis utilizes two datasets:

- `DA441.csv`: Daily return data for 441 U.S. companies listed on the S&P 500 from 1 January 2000 to 31 December 2023.
- `RiskFree.csv`: Daily U.S. risk-free rate (expressed in percentage terms), obtained from the Kenneth French Data Library.[1]

Both datasets are assumed to be synchronized by date and pre-cleaned to eliminate missing values. This transformation is applied to each observation.

The risk-free rate is provided in percentage form (e.g., 0.05 represents 5%). These values are converted into decimal format as shown below:

$$r_{f,t}^{(\text{decimal})} = \frac{r_{f,t}^{(\text{percent})}}{100}$$

Figure 1 shows the evolution of the U.S. risk-free rate over the full period 2000–2023.

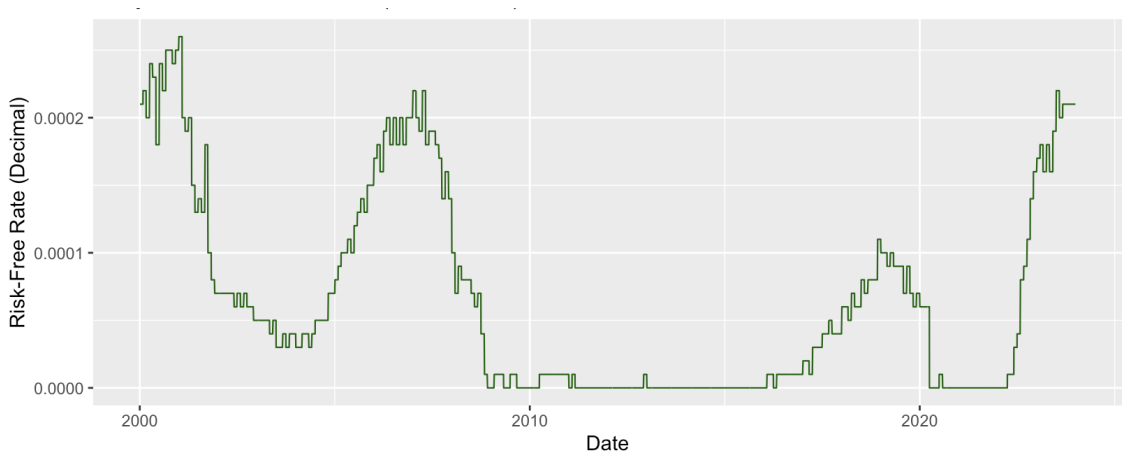



**Figure 1:** Risk-free rate over the full period (2000-2023)

To reduce dimensionality and simulate practical portfolio construction, a random subset of 100 companies is selected from the full list of 441. This selection is performed using a fixed random seed corresponding to the group number. This ensures the result is reproducible and unique to each group. In our case, Group 10, we apply:

```r
set.seed(10)
selected_columns <- sample(colnames(asset_data)[2:442], 100)
```

**Listing 1:** Random selection of 100 companies using Group 10 seed

The daily excess return for company $i$ on day $t$, denoted $r_{i,t}^{\text{excess}}$, is defined by Formula 1:

$$r_{i,t}^{\text{excess}} = r_{i,t} - r_{f,t} \tag{1}$$

where $r_{i,t}$ is the raw daily return of stock $i$, and $r_{f,t}$ is the corresponding daily risk-free rate. This operation is vectorized across all 100 selected stocks and all trading days within the group-specific training and testing windows.

[1] https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

To better understand the distributional properties of the selected equities, we compute descriptive statistics for each asset's raw daily return over the analysis period from 26 February 2020 to 29 December 2023.

Table 1 shows the top 10 companies ranked by annualised return, along with their corresponding annualised volatility (standard deviation), skewness, and kurtosis. Notably, several high-return stocks—such as NVDA and RIG—also display elevated skewness and extreme kurtosis, indicative of fat-tailed distributions and asymmetric return profiles.

**Table 1:** Top 10 Companies by Annualised Return (2020–2023)

| Ticker | Mean Return | Volatility (SD) | Skewness | Kurtosis |
|---|---|---|---|---|
| NVDA | 0.6748 | 0.5453 | 0.4347 | 4.3310 |
| RIG | 0.6317 | 0.9911 | 1.0038 | 12.0769 |
| SWN | 0.5887 | 0.6454 | 0.5683 | 2.1514 |
| FCX | 0.5178 | 0.5484 | 0.4239 | 7.2125 |
| LB | 0.4436 | 0.6519 | 1.5781 | 18.2225 |
| OXY | 0.4087 | 0.6976 | -0.6744 | 23.4630 |
| AMAT | 0.3890 | 0.4806 | -0.2946 | 3.5939 |
| HAL | 0.3756 | 0.6101 | -0.8399 | 15.1502 |
| RCL | 0.3702 | 0.7344 | 0.1309 | 8.3229 |
| LPX | 0.3549 | 0.5132 | 0.1005 | 8.4566 |

*Note:* The Return and volatility figures are annualised. Skewness and kurtosis are computed on daily returns.

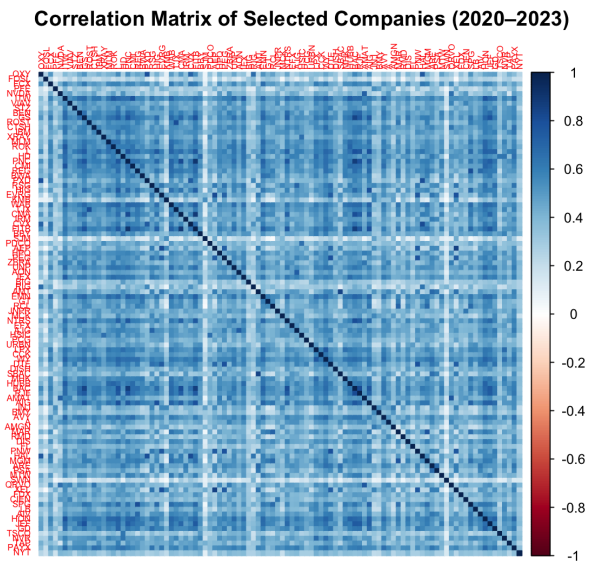



**Figure 2:** Correlation Matrix of Selected U.S. Equities (2020–2023)

To assess the degree of linear dependence between assets, we compute the pairwise Pearson correlation matrix using daily raw returns from February 2020 to December 2023. This allows us to examine the structural comovement of the selected 100 U.S. equities during a period of significant market stress and recovery.

**Figure 2** displays the resulting heatmap of correlations. The matrix reveals a dominant presence of strong positive relationships across most assets, suggesting shared exposure to systematic risk factors such as market momentum, sector cycles, and macroeconomic shocks. Relatively few pairs exhibit weak or negative correlations, indicating limited opportunities for pure diversification.

## A2: Selection of Estimation Models

The analysis covers February 2020 to December 2023, a period marked by major financial events: the COVID-19 pandemic and global market crash (2020), the post-pandemic rally (2021), the Russia–Ukraine war and energy crisis (2022), and persistent inflation with rising interest rates (2022–2023). Estimators are chosen to reduce the influence of such shocks, ensuring stable portfolio decisions under extreme market conditions.

Based on empirical performance, statistical robustness, and theoretical justification under high-dimensional conditions ($p > T$), three estimator combinations are selected for further evaluation as shown below.

### Classical Approach: Sample Mean and Sample Covariance (Sample-Sample)

It was used as a benchmark combination. It represents the unregularised baseline, applying empirical return and risk estimates directly from the data. While theoretically unbiased, this estimator is particularly unstable in short windows and sensitive to extreme observations. Given that some training windows include periods of acute market stress, such as March 2020's COVID-19 crash and the 2022 inflation spike, this approach often yields erratic inputs that propagate through the optimisation process.

### D-Multi-Shrinkage Mean with Ledoit-Wolf Covariance Estimator (DM–cov1Para)

The second combination strikes a more robust balance by incorporating temporal structure and covariance regularisation. The Dynamic Mean gives greater weight to recent returns, allowing the estimator to react more quickly to shifting return dynamics. This proved especially relevant during the volatile periods of mid-2020 and late 2022, when directional trends shifted sharply. In parallel, the cov1Para estimator stabilises risk estimation by shrinking the sample covariance matrix toward a structured target. This method was well suited for addressing the instability observed in Sample-Sample and allowed for more consistent portfolio weights across rebalancing windows.

### Oracle Linear Shrinkage Mean with Quasi-Inverse Shrinkage Covariance Estimator (OL–QIS)

Lastly, the combination aimed to explore the benefits of more advanced regularisation. OL shrinks the sample mean vector toward a theoretically optimal target derived under a quadratic loss framework. QIS, meanwhile, applies nonlinear shrinkage to the eigenvalues of the covariance matrix, regularising risk estimates in a way that preserves key distributional features of the original data. This combination is particularly relevant in a high-$p$, low-$T$ setting where inversion of the raw covariance matrix can be numerically unstable. Despite its theoretical appeal, this pairing showed mixed performance, possibly due to its sensitivity to the extreme return shocks seen in some windows, especially in late 2022 and early 2023, when macro conditions remained volatile.

Overall, the chosen combinations reflect a cross-section of modelling philosophies, from empirical to fully regularised, and were selected to test not only performance but also the resilience of parameter estimates in rapidly evolving financial markets.

## A3: Portfolio Construction and Rolling Window Setup

For Group 10, the initial training dataset encompasses the entirety of Period 10, spanning from February 26, 2020 to January 24, 2022. The corresponding initial testing dataset covers the subsequent six months within Period 11.

The rolling window procedure proceeds by incrementally expanding the training dataset: the second training dataset includes the initial training period plus the first month of Period 11. Correspondingly, the second testing dataset consists of the six months immediately following this expanded training period.

This incremental process continues with a fixed one-month step forward (approximately 21 trading days), where each iteration's training set grows by one month, and the testing set is a fixed six-month window immediately following the training period. This rolling evaluation continues until the final testing dataset concludes at the end of Period 11, December 29, 2023. For the given data, we have 18 total segments.

The R output and the summary table containing all relevant dates are presented below:

```
Total segments: 18
Training window expands from 1 to 358 days
Fixed testing window: 126 days (6.0 months)

First Evaluation:
Training: 2020-02-26 to 2020-02-26
Testing:  2022-01-24 to 2022-07-25

Final Evaluation:
Training: 2020-02-26 to 2021-07-27
Testing:  2022-01-24 to 2022-07-25
```

**Table 2:** Summary of Training and Testing Periods for Each Rolling Window

| Window | Training Period | Testing Period |
|---|---|---|
| 1 | 2020-02-26 to 2022-01-24 | 2022-01-24 to 2022-07-25 |
| 2 | 2020-02-26 to 2022-02-23 | 2022-02-23 to 2022-08-23 |
| 3 | 2020-02-26 to 2022-03-24 | 2022-03-24 to 2022-09-22 |
| 4 | 2020-02-26 to 2022-04-25 | 2022-04-25 to 2022-10-21 |
| 5 | 2020-02-26 to 2022-05-24 | 2022-05-24 to 2022-11-21 |
| 6 | 2020-02-26 to 2022-06-24 | 2022-06-24 to 2022-12-21 |
| 7 | 2020-02-26 to 2022-07-26 | 2022-07-26 to 2023-01-24 |
| 8 | 2020-02-26 to 2022-08-24 | 2022-08-24 to 2023-02-23 |
| 9 | 2020-02-26 to 2022-09-23 | 2022-09-23 to 2023-03-24 |
| 10 | 2020-02-26 to 2022-10-24 | 2022-10-24 to 2023-04-25 |
| 11 | 2020-02-26 to 2022-11-22 | 2022-11-22 to 2023-05-24 |
| 12 | 2020-02-26 to 2022-12-22 | 2022-12-22 to 2023-06-26 |
| 13 | 2020-02-26 to 2023-01-25 | 2023-01-25 to 2023-07-26 |
| 14 | 2020-02-26 to 2023-02-24 | 2023-02-24 to 2023-08-24 |
| 15 | 2020-02-26 to 2023-03-27 | 2023-03-27 to 2023-09-25 |
| 16 | 2020-02-26 to 2023-04-26 | 2023-04-26 to 2023-10-24 |
| 17 | 2020-02-26 to 2023-05-25 | 2023-05-25 to 2023-11-22 |
| 18 | 2020-02-26 to 2023-06-27 | 2023-06-27 to 2023-12-22 |

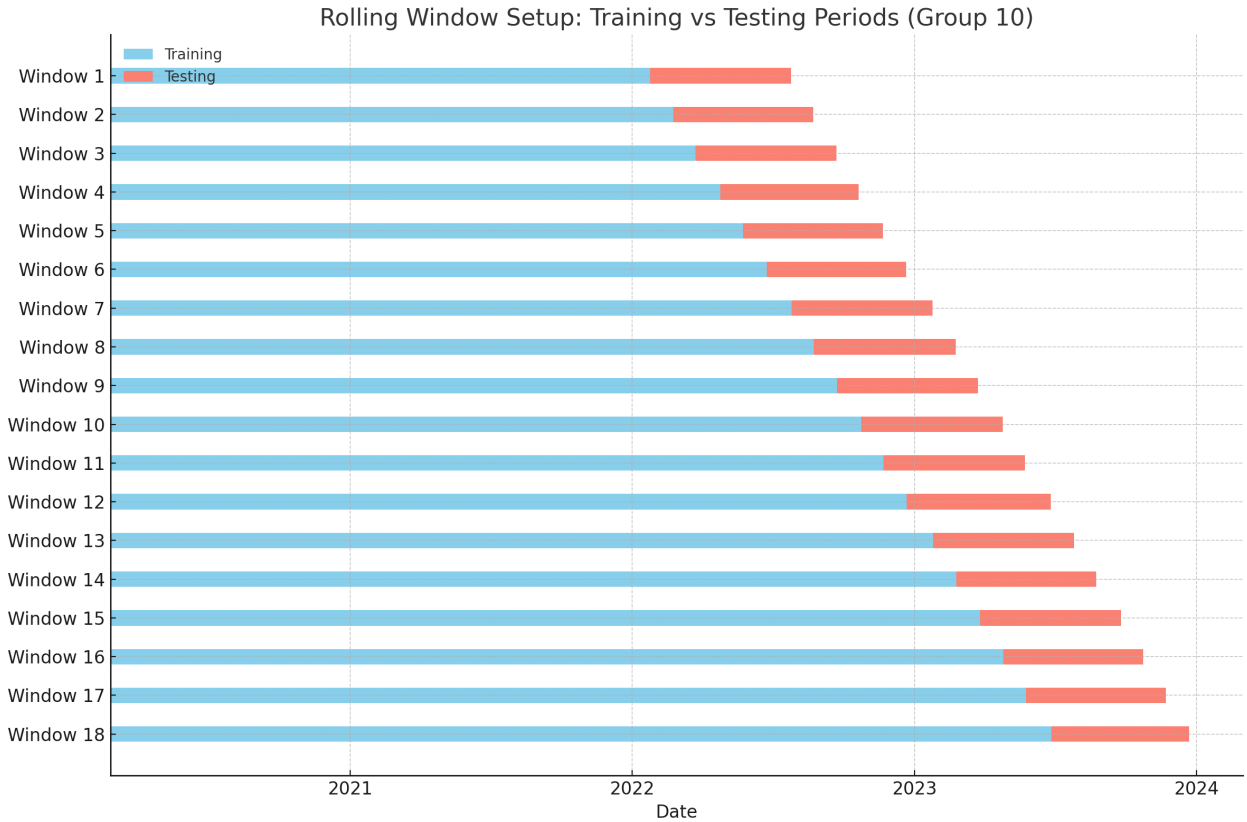



**Figure 3:** Rolling window setup diagram

As shown in Figure 3, each evaluation window begins with a training set that expands forward in time, followed by a fixed-length testing period of approximately 6 months. This structure ensures robustness and consistency across varying market conditions.

## A4: Portfolio Performance Evaluation

For each testing window and each of the four portfolios (three tangency portfolios based on the selected estimation models, plus an equal weight benchmark), we evaluate portfolio performance using the following metrics:

### Annualised Sharpe Ratios by Rolling Window

The Sharpe ratio measures the risk-adjusted return, computed as the ratio of the mean excess return to the standard deviation of excess returns, annualised to account for the number of trading days per year. It is given by:

$$\text{Sharpe Ratio}_{annual} = \frac{\bar{r}_p}{\sigma_p} \times \sqrt{252}$$

where:

- $\bar{r}_p$ is the mean daily excess return of the portfolio over the testing window,
- $\sigma_p$ is the standard deviation of daily excess returns of the portfolio over the testing window.

We use the `SharpeRatio.annualized` function from the `PerformanceAnalytics` R package to compute this value for each portfolio and testing period. The annualised ratio for each window is given in the table below.

**Table 3:** Annualised Sharpe Ratios by Evaluation Window

| Window | Sample-Sample | cov1Para-DM | qis-OL | EqualWeight |
|---|---|---|---|---|
| 1 | -0.916 | -0.848 | -0.539 | -0.549 |
| 2 | -0.749 | -0.641 | -0.366 | -0.089 |
| 3 | -1.105 | -1.115 | -1.173 | -1.153 |
| 4 | -0.383 | -0.138 | -0.844 | -0.908 |
| 5 | 0.608 | 1.037 | 0.498 | 0.337 |
| 6 | 0.200 | 0.874 | 0.016 | 0.833 |
| 7 | -0.588 | 0.106 | -0.790 | 0.828 |
| 8 | -1.260 | -0.770 | -1.360 | 0.300 |
| 9 | -1.343 | -1.184 | -0.864 | 0.499 |
| 10 | -0.877 | -0.641 | -0.347 | 0.745 |
| 11 | -0.363 | -0.281 | -0.532 | -0.555 |
| 12 | -0.479 | 0.149 | -0.404 | 0.175 |
| 13 | 0.630 | 0.979 | 0.847 | 0.420 |
| 14 | 1.853 | 2.420 | 1.721 | -0.275 |
| 15 | 2.229 | 2.671 | 1.589 | 0.358 |
| 16 | 1.435 | 2.019 | 0.461 | -0.753 |
| 17 | 1.423 | 1.815 | 1.015 | 0.414 |
| 18 | 0.137 | 0.128 | -0.297 | 0.910 |

### Nominal Gains Ratios by Rolling Window

Nominal gain is the absolute monetary profit or loss generated by the portfolio over the testing period, assuming an initial investment amount $I_0$. It is computed as:

$$\text{Nominal Gain} = I_0 \times \left( \prod_{t=1}^{N} (1 + r_{p,t}) - 1 \right)$$

where:

- $r_{p,t}$ is the portfolio return on day $t$ in the testing window,
- $N$ is the number of days in the testing window (fixed at 126 days),
- $I_0$ is the initial investment (£1,000,000 in our case).

This cumulative return calculation assumes geometric compounding of daily returns. The nominal ratio for each window is given in the table below.

**Table 4:** Nominal Profits by Evaluation Window (£1,000,000 Initial Investment)

| Window | Sample-Sample | cov1Para-DM | qis-OL | EqualWeight |
|---|---|---|---|---|
| 1 | -£324,006 | -£420,339 | -£89,656 | -£77,888 |
| 2 | -£324,231 | -£449,835 | -£73,241 | -£25,225 |
| 3 | -£407,398 | -£554,101 | -£169,984 | -£144,550 |
| 4 | -£209,846 | -£236,979 | -£110,910 | -£129,298 |
| 5 | £86,881 | £301,599 | £43,394 | £27,578 |
| 6 | -£60,639 | £210,543 | -£8,065 | £91,838 |
| 7 | -£328,675 | -£185,744 | -£83,334 | £90,258 |
| 8 | -£510,312 | -£535,208 | -£126,758 | £22,090 |
| 9 | -£680,449 | -£715,571 | -£105,945 | £46,055 |
| 10 | -£592,002 | -£590,204 | -£59,656 | £67,971 |
| 11 | -£292,714 | -£354,279 | -£58,877 | -£56,110 |
| 12 | -£361,696 | -£187,191 | -£47,812 | £7,753 |
| 13 | £81,231 | £178,299 | £75,566 | £27,980 |
| 14 | £881,717 | £2,623,748 | £173,369 | -£26,918 |
| 15 | £1,183,862 | £2,517,832 | £172,363 | £19,948 |
| 16 | £547,998 | £1,209,764 | £41,269 | -£55,807 |
| 17 | £642,219 | £1,256,211 | £142,538 | £25,257 |
| 18 | -£215,910 | -£374,475 | -£61,287 | £63,414 |

### Summary Statistics of Annualised Sharpe Ratios

Table 5 presents the summary statistics for the annualised Sharpe ratios of each portfolio strategy across the rolling evaluation windows. For each strategy, the Sharpe ratio was computed for every out-of-sample testing period, capturing the risk-adjusted return performance as shown above.

The mean provides the average Sharpe ratio over all rolling windows, reflecting the typical risk-return trade-off of the strategy. The median offers a robust central tendency measure less affected by extreme values. The standard deviation quantifies the variability in the Sharpe ratios, indicating the consistency of the strategy's performance over time.

These statistics together enable a comprehensive comparison of portfolio effectiveness and stability among the evaluated strategies.

**Table 5:** Summary Statistics of Annualised Sharpe Ratios

| Strategy | Mean | Median | SD |
|---|---|---|---|
| cov1Para-DM (Combination 2) | 0.3655 | 0.1170 | 1.2290 |
| EqualWeight | 0.0855 | 0.3186 | 0.6419 |
| Sample-Sample (Combination 1) | 0.0252 | -0.3728 | 1.1097 |
| qis-OL (Combination 3) | -0.0761 | -0.3568 | 0.9068 |

### Summary Statistics of Nominal Gains (£)

Table 6 reports the summary statistics for the nominal gains achieved by each portfolio strategy over the rolling testing periods. The nominal gain is calculated as the absolute profit or loss in pounds (£) on an initial investment of £1,000,000 during each out-of-sample test window.

The mean nominal gain represents the average monetary return across all rolling windows, providing insight into the typical profitability of each strategy. The median gain offers a measure of central tendency less sensitive to outliers. The standard deviation indicates the volatility of nominal returns, highlighting the variability in performance over time.

These statistics collectively facilitate a comparative assessment of the financial outcomes and risk exposure associated with each portfolio approach.

**Table 6:** Summary Statistics of Nominal Gains (£)

| Strategy | Mean (£) | Median (£) | SD (£) |
|---|---|---|---|
| cov1Para-DM (Combination 2) | 205,226.08 | -212,085.29 | 1,026,084.3 |
| EqualWeight | -1,425.12 | 21,018.79 | 69,340.0 |
| qis-OL (Combination 3) | -19,279.18 | -59,266.28 | 104,221.4 |
| Sample-Sample (Combination 1) | -49,109.44 | -254,312.02 | 527,551.1 |

Figures 4 and 5 provide a graphical representation of each strategy's performance across the 18 evaluation windows.

**Figure 4** illustrates the nominal profit or loss (in £ millions) per strategy over each six-month testing window. Combination 2 (cov1Para-DM) consistently outperforms the other strategies, delivering substantial gains in certain windows (e.g., Windows 14 and 15). By contrast, Combinations 1 and 3 generate more volatile and often negative returns.

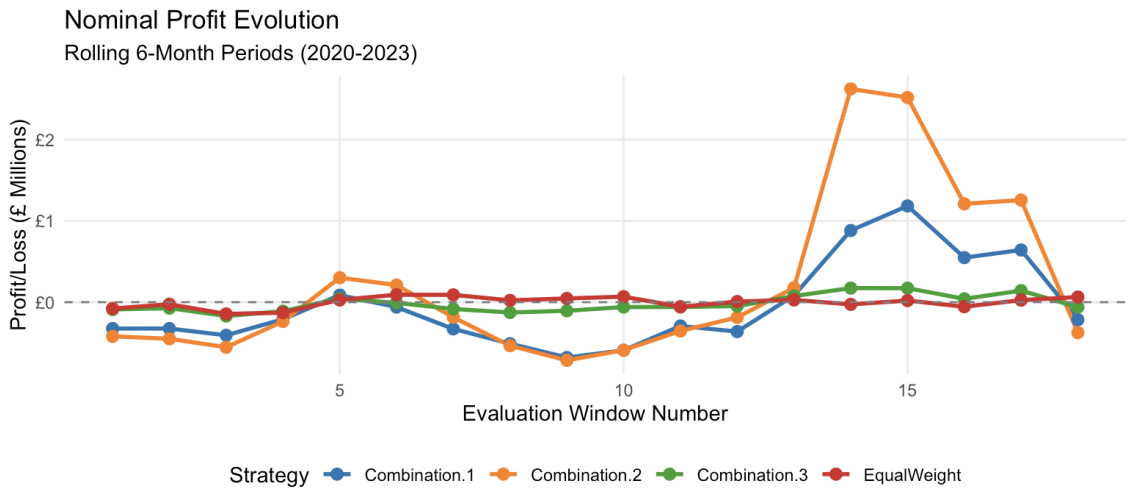



**Figure 4:** Nominal Profit evolution for each window

**Figure 5** shows the evolution of annualised Sharpe ratios across windows. Again, Combination 2 demonstrates superior risk-adjusted returns, particularly in mid-to-late 2022 and 2023. Combination 3 (qis-OL) remains conservative but delivers relatively low Sharpe ratios, suggesting an overly cautious risk allocation.

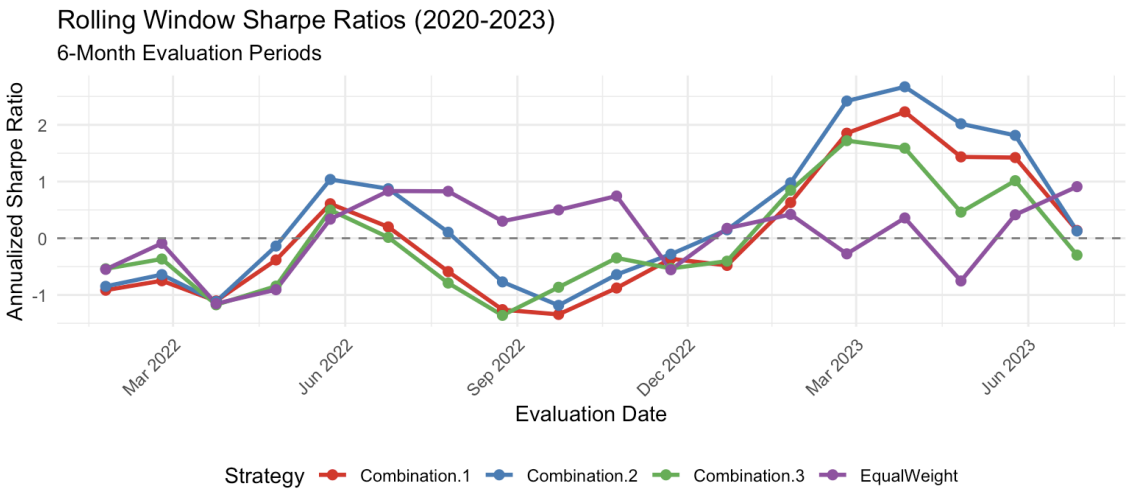



**Figure 5:** Sharpe ratio evolution for each window

These plots highlight the robustness and consistency of the shrinkage-based estimators used in Combination 2 under varying market conditions.

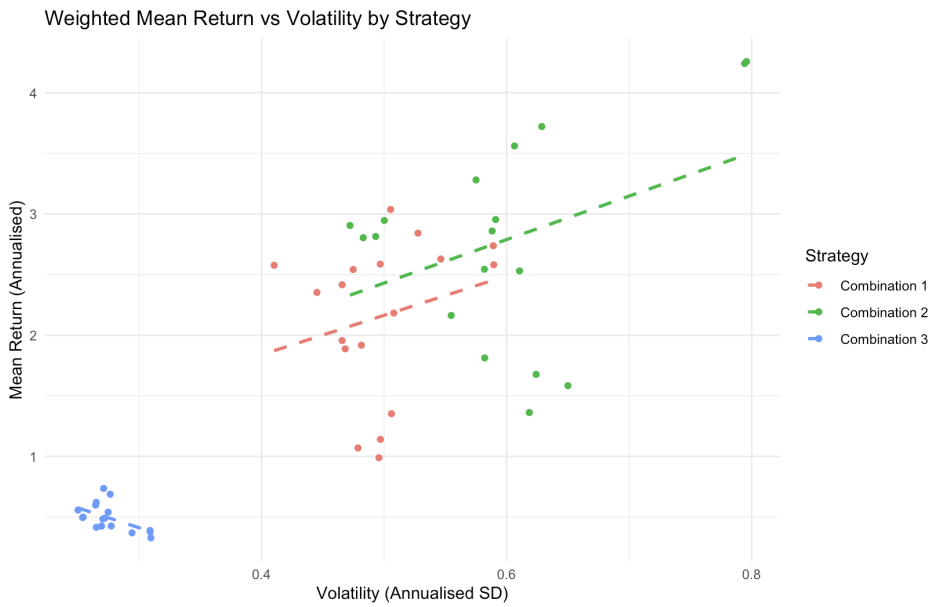



**Figure 6:** Portfolio-Weighted Annualised Return vs Volatility by Strategy across Rolling Windows

Figure 6 highlights clear differences in portfolio behavior across strategies. Combination 2 (green) occupies the high-return, high-volatility region, indicating a risk-seeking allocation with exposure to high-growth assets. In contrast, Combination 3 (blue) is tightly clustered in the low-return, low-volatility corner, suggesting an overly conservative allocation likely driven by excessive regularisation. Combination 1 (red) shows a more balanced distribution, reflecting moderate exposure across the return-risk spectrum. These results underscore how shrinkage specification affects both the aggressiveness and selectivity of the portfolio.

## A5: Interpretation of Results and Executive Summary

This section synthesises the key insights drawn from the simulation results, with particular emphasis on how estimation methods affected portfolio performance across changing market conditions. The analysis draws on both Sharpe ratios and nominal returns, reflecting both risk-adjusted and absolute profitability measures. These metrics were tracked across 18 rolling windows, each representing different macroeconomic regimes.

**Table 7:** Summary Statistics of Strategy Performance: Sharpe Ratios and Nominal Gains (£)

| Portfolio Combination | Mean | Median | SD |
|---|---|---|---|
| **Sharpe Ratios (Annualised)** | | | |
| Sample-Sample (Combination 1) | 0.0252 | -0.3728 | 1.1097 |
| cov1Para-DM (Combination 2) | 0.3655 | 0.1170 | 1.2290 |
| qis-OL (Combination 3) | -0.0761 | -0.3568 | 0.9068 |
| EqualWeight | 0.0855 | 0.3186 | 0.6419 |
| **Nominal Gains (£1,000,000 Initial Investment)** | | | |
| Sample-Sample (Combination 1) | -49,109.44 | -254,312.02 | 527,551.1 |
| cov1Para-DM (Combination 2) | 205,226.08 | -212,085.29 | 1,026,084.3 |
| qis-OL (Combination 3) | -19,279.18 | -59,266.28 | 104,221.4 |
| EqualWeight | -1,425.12 | 21,018.79 | 69,340.0 |

### Risk-Adjusted Performance (Sharpe Ratios)

The cov1Para–DM combination clearly led the group, posting the highest average annualised Sharpe ratio of 0.3655. This result highlights the strength of combining time-sensitive expected returns with regularised risk estimates. The method demonstrated adaptability during turbulent periods such as mid-2022, when inflation uncertainty and rate hikes disrupted markets. Although Equal Weight achieved a modest mean Sharpe ratio of 0.0855, it was more stable across windows due to its simplicity and lack of reliance on noisy parameter estimates. In contrast, both Sample-Sample and OL–QIS showed more erratic performance, with negative average Sharpe ratios and higher volatility across windows. These combinations struggled during high-stress periods, underlining the limitations of unregularised or overly complex estimators in live portfolio settings.

### Absolute Performance (Nominal Returns)

The cov1Para–DM portfolio also dominated in nominal terms, delivering an average gain of £205,226.08 per window. These gains were particularly concentrated during bullish recoveries such as early 2023, when equities rebounded on expectations of easing inflation. However, the high standard deviation and negative median suggest that the strategy is also exposed to substantial downside during shocks, like those seen during the energy crisis or the initial COVID collapse. Equal Weight again proved consistent, if unspectacular, with a near-zero average gain and lower volatility. Sample-Sample and OL–QIS portfolios underperformed, with negative means and wide dispersion in returns.

### Contextual Interpretation

The rolling nature of the analysis captures different financial regimes, from the tail end of COVID-induced volatility through the 2022 inflation crisis and into a more stabilised 2023 macro backdrop. The strong outperformance of cov1Para–DM during windows that coincided with economic recoveries and monetary tightening suggests that this combination is

responsive enough to benefit from directional shifts, while regularised enough to avoid over-reacting to short-term noise.

Conversely, Sample-Sample suffered in nearly all windows, reaffirming that naïve estimators often collapse under structural breaks and elevated market uncertainty. OL–QIS, while theoretically robust, appeared to be too sensitive to estimation error in the limited-window setup, particularly when extreme returns skewed eigenvalue distributions.

### Final Takeaway

These results indicate that using regularised estimators like cov1Para–DM and StM–QIS can offer meaningful advantages in real-world portfolio construction. They allow the investor to capitalise on changing return environments without amplifying risk through noisy parameter inputs. While Equal Weight offers a safety net during extreme uncertainty, it leaves returns on the table when market timing and risk structure can be exploited. This analysis supports the broader argument that combining shrinkage techniques with adaptive return models results in portfolios that are not only theoretically optimal but empirically resilient.

# Coursework 2: Shrinkage Generalised Linear Models for Insurance Claims Estimation

## B1: In-Sample Performance Evaluation

### Objective

The aim of this analysis is to evaluate the in-sample performance of five Generalised Linear Models (GLMs): OLS, SR, GSR, St, and DSh, using the Tweedie family with a log link function. Four power parameters ($p \in \{1.25, 1.5, 1.75, 2\}$) are tested, and model performance is assessed using the Mean Squared Error (MSE).

The Tweedie family is chosen because it models positive, right-skewed data, with variance following:

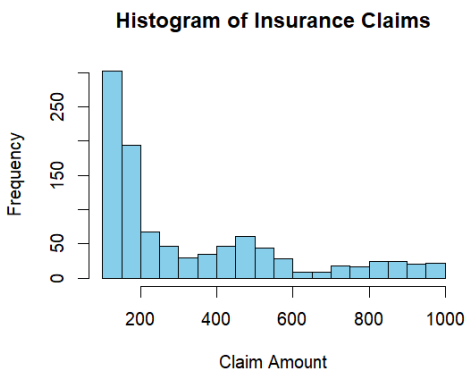
$$\mathrm{Var}(Y) = \phi\mu^p,$$

where $\phi > 0$ is a dispersion parameter and $p$ is the power parameter. Lower values of $p$ resemble compound Poisson–Gamma distributions, while higher values behave more like Gamma, making Tweedie suitable for heterogeneous claims.

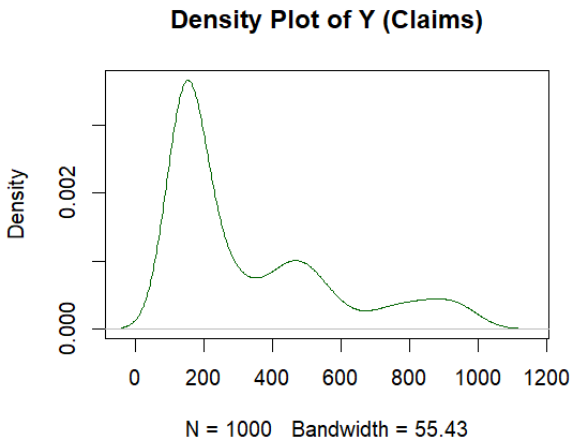
### Data Summary and Visualisation

The dataset contains 1,000 insurance observations with 50 covariates: the first 30 are binary indicators, and the remaining 20 are continuous. The response variable $Y$ represents positive insurance claims.

The claims have a mean of approximately 333, a standard deviation of 245, and skewness of 1.20, indicating a right-skewed distribution typical for insurance data.




**(a)** Histogram of claims




**(b)** Density plot of claims

**Figure 7:** Distribution of insurance claims $Y$: heavy right-skew with long tail.

**Table 8:** Summary statistics for insurance claims $Y$

| Statistic | Value |
|---|---|
| Mean | 332.59 |
| Std. Dev | 245.20 |
| Skewness | 1.20 |
| Min | 127.48 |
| 1st Quartile | 145.00 |
| Median | 202.50 |
| 3rd Quartile | 474.90 |
| Max | 993.66 |

### Methodology

We fitted five Generalised Linear Models (GLMs) using the Tweedie family with a log link function for four different power parameters $p \in \{1.25, 1.5, 1.75, 2\}$. The models considered were:

- **OLS**: Classical GLM without shrinkage.
- **SR**: Shrinkage Ridge (L2 penalty).
- **GSR**: Group Shrinkage Ridge, which applies penalties by groups of variables.
- **St**: Lasso (L1 penalty).
- **DSh**: Horseshoe prior, an adaptive Bayesian shrinkage method.

The log link was used in all models to ensure positive predictions and to linearise multiplicative effects of predictors. The augmented design matrix included an intercept term, resulting in 51 covariates (50 features + intercept).

Predicted mean claims were computed using:

$$\hat{y}_i = \exp(\tilde{x}_i^\top \hat{\beta}),$$

where:

- $\hat{y}_i$ is the predicted mean claim for observation $i$,
- $\tilde{x}_i \in \mathbb{R}^{51}$ is the augmented covariate vector for observation $i$, including a leading 1 for the intercept,
- $\hat{\beta}$ is the estimated coefficient vector for the corresponding GLM.

This formulation applies the log link and ensures strictly positive predictions.

To assess in-sample performance, we computed the Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2,$$

where $n = 1000$ is the number of observations, $y_i$ is the observed claim, and $\hat{y}_i$ is the predicted claim. This was calculated for all 20 combinations of model and power parameter.

### Results

| Model | $p = 1.25$ | $p = 1.5$ | $p = 1.75$ | $p = 2.0$ |
|---|---|---|---|---|
| OLS | 593.80 | 608.95 | 627.72 | 650.10 |
| SR | 593.79 | 608.94 | 627.74 | 650.06 |
| GSR | **593.03** | 608.99 | 626.51 | 648.90 |
| St | 593.85 | 609.03 | 627.88 | 650.44 |
| DSh | 598.31 | **606.90** | **620.43** | **641.04** |

**Table 9:** In-sample MSE for each model and Tweedie power parameter

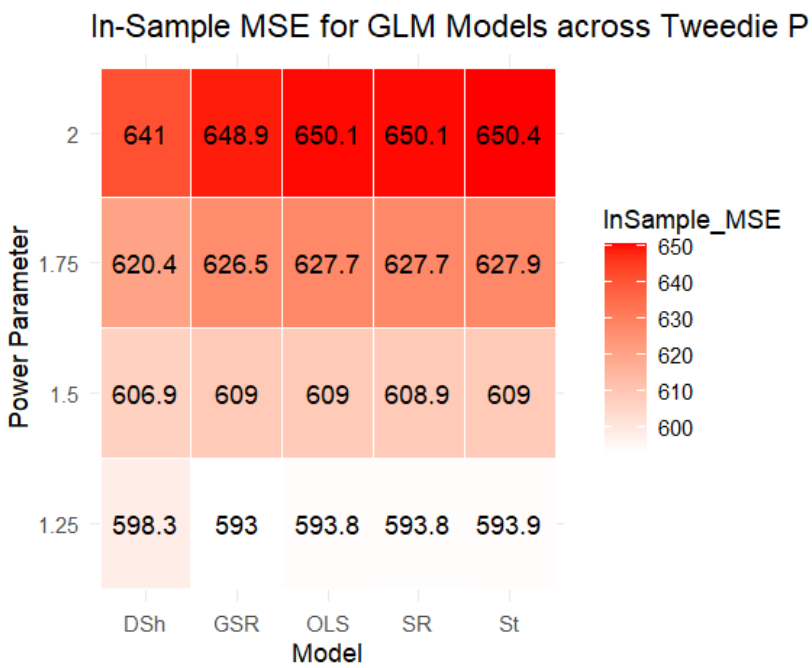



**Figure 8:** Heatmap of in-sample MSE across models and power parameters.

DSh achieves the lowest MSE for $p = 1.5, 1.75$, and 2.0, while GSR performs best at $p = 1.25$. OLS consistently has the highest MSE, confirming that shrinkage improves accuracy. MSE increases as $p$ increases, reflecting greater variance in Tweedie models.

### Interpretation

For $p = 1.25$, GSR achieves the lowest MSE (593.03). For $p = 1.5, 1.75$, and 2.0, DSh achieves the lowest MSEs of 606.90, 620.43, and 641.04 respectively.

Shrinkage-based models, particularly GSR and DSh, consistently outperform OLS. The performance gap is small at lower power values but becomes more pronounced as the power parameter increases. GSR performs best at low variance (small $p$) by leveraging group structure effectively, while DSh dominates at higher powers, where its adaptive shrinkage (via the horseshoe prior) better handles large variance and noise.

OLS and SR deliver very similar results, indicating that ridge shrinkage adds only minimal improvement in this context. All models show increasing MSE as $p$ increases, reflecting the Tweedie variance structure:

$$\mathrm{Var}(Y) \propto \mu^p,$$

so higher $p$ corresponds to greater variability and more difficult prediction.

These findings align with theoretical expectations: regularisation stabilises coefficient estimates and mitigates overfitting, an effect that becomes increasingly important as data variance grows. Overall, these results support the use of shrinkage-based GLMs for high-dimensional, skewed insurance data. We now perform a similar evaluation of out-of-sample performance.

## B2: Out-of-Sample Performance Evaluation

To assess how well each estimation method generalised to out-of-sample data, we split the 1,000 policies into a 70/30 training/test split 100 times. In each split, and for each of the four Tweedie power parameters, we fitted the five GLMs using a log link function. Next, we generated predictions on the 30% test set and computed the average mean square error (MSE) from all the 100 test MSEs. Finally, we measured how often each model achieved the lowest MSE within each power parameter.

| Power | Model | AvgTestMSE | LowestCount |
|---|---|---|---|
| 1.25 | OLS | 716.1129 | 2 |
| 1.25 | SR | 716.0834 | 3 |
| 1.25 | GSR | 716.8043 | 15 |
| 1.25 | St | 716.1630 | 1 |
| 1.25 | DSh | 703.4849 | 79 |
| 1.5 | OLS | 721.7608 | 1 |
| 1.5 | SR | 721.7284 | 6 |
| 1.5 | GSR | 722.2278 | 8 |
| 1.5 | St | 721.8627 | 4 |
| 1.5 | DSh | 707.4958 | 81 |
| 1.75 | OLS | 732.0587 | 1 |
| 1.75 | SR | 732.0333 | 6 |
| 1.75 | GSR | 732.6140 | 9 |
| 1.75 | St | 732.2744 | 3 |
| 1.75 | DSh | 716.1000 | 81 |
| 2 | OLS | 747.1528 | 2 |
| 2 | SR | 747.1018 | 5 |
| 2 | GSR | 747.6534 | 9 |
| 2 | St | 747.5523 | 4 |
| 2 | DSh | 730.7598 | 80 |

**Table 10:** Model performance

The results in Table 10 show that, for each power parameter out of 100 test splits, DSh had the largest quantity of lowest MSE counts by a substantial margin. This implies that DSh was the best-performing model out-of-sample due to its stronger regularisation, which shrinks the GLM more aggressively, thereby protecting the model against over-fitting and resulting in lower MSE on new data.

In contrast, the simple OLS model had the highest average test MSE, which is a clear sign of its tendency to over-fit the training data. The single-penalty ridge methods (SR and GSR), and the stabilised variant (St), had highly similar MSEs to OLS, which indicates their limitations in improving model fit.

It can be seen that the choice of power parameter affects out-of-sample performance by creating heavier tails, which lead to higher errors. This is because, as the power grows, the

variance of the distribution grows by that power, thereby resulting in more extreme values. DSh's average test MSE rises from roughly 707 to 733, while OLS's average MSE similarly rises from roughly 722 to 751.

The rankings between in-sample and out-of-sample performance are partly consistent. Both analyses show that MSE for all models increases with the power parameter, and that shrinkage models generally outperform OLS. In-sample, shrinkage methods generally outperform OLS, especially at higher powers, whereas out-of-sample, OLS has the highest average MSE across all powers. In-sample, GSR has the lowest MSE at $p = 1.25$, while DSh has the lowest for all higher powers. Out-of-sample, DSh dominates across all powers. GSR fits training data well at low variance in-sample. Out-of-sample, DSh's adaptive shrinkage gives stronger regularisation, preventing over-fitting when variance grows at higher powers.